THE HONG KONG
UNIVERSITY OF SCIENCE AND
TECHNOLOGY (GUANGZHOU)

Relation Construction for Battery-Free Design An Actual Attempt and Post-Hoc Validation

Development manuscript

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Abstract

A model-generated relation can affect scientific design only if its source, interpretation, qualification and consumer use survive the host interface. We report one actual battery-free design episode and a separately authorized post-hoc adapter evaluation. The model acquired three sources and constructed an energy comparison, but invalid JSON Pointer paths exhausted the registered one-proposal allowance. The original episode therefore established no successful E5 chain. A subsequent evaluation reused the model's already generated corrected call, with no new model proposal or provider request. Source binding, independent units and Decimal evaluation qualified a source-dependent relation; a real consumer used the original policy to change capacitance from 10 to 15 μF. The independently checked nominal service changed from FAIL to PASS. A strong static complete-information design also passed with 10 μF and earlier transmission. The result establishes a post-hoc engineering connection from constructed relation to executed design, while the original failure, static baseline, unknown physical qualification and absence of autonomous discovery remain explicit.

1 Question and evidence scope

Battery-free service depends on the ordering of energy arrival, storage, startup, event capture, reset and transmission. A derived energy comparison can summarize part of this behavior, but a design decision still needs an executable input edge and a complete service check. We examine whether an actual model-generated observation can enter that chain without replacing a failed attempt with a retrospectively improved outcome.

The unit of evidence is the public development episode `p2-e5-001` and its separately labeled adapter evaluation. The original episode result is retained as a syntax rejection [1]. The post-hoc receipts bind the previously generated corrected call to external qualification, an executed consumer and an independent service check [2, 3, 4]. These are two evidence records with different admissibility conditions. They do not form a successful preregistered live episode or a population-level comparison.

2 Bounded proposal and external authority

The host exposes source listing, acquisition, proposal and submission. A proposal declares a hypothesis, source identifier, units and domain. Selection names an existing JSON field. Construction supplies a bounded JSON abstract syntax tree (AST) over fields, constants and permitted operators. A construction must contain a source field and an operator; this mode label establishes no novelty. The initial schema supplies no answer expression, required-field list or target relation. The expression below reports actual model output retrospectively.

The frozen B interpreter accepts at most 64 AST nodes and depth eight, an 8 MB JSON source and 32,768 vector elements. JSON Pointer paths identify data; arbitrary Python and filesystem access are unavailable through the AST. Unit dimensions must agree across operations, output units must match the root and declared numeric domain bounds are checked on source values. Missing values, invalid arithmetic or an unsupported domain yield UNKNOWN or a retained rejection. These gates address bounded interpretation, not physical validity.

The two-argument evaluator accepts the corresponding raw JSON payload. With no external field-unit map it reports declared units only. An S0-supplied map can check unit consistency, while source authenticity and independent qualification remain outside the proposer. The host separately binds original source bytes, the model call, actual cost and consumer input. An OBSERVED value from the B interpreter cannot certify its own source or service outcome.

The model also proposes a conditional design table with at most three branches. Each branch contains a strictly typed equality value and a legal capacitance and earliest-transmission design. The generic consumer starts with UNKNOWN when no qualified observation exists. After external qualification it matches the observed value to the original table and passes the selected full input to the BF service. No expected service label is needed by the matching rule.

3 The original actual episode failed

The model acquired `component_record`, `excitation_record` and `saved_activity`. These supplied the device record, input/event/disturbance record and saved actions. It then attempted a construction over those acquired facts. Its first proposal omitted the leading slash required by JSON Pointer. The host rejected it with `JSON_POINTER_REQUIRED`; that attempt consumed the registered one-proposal allowance. The model subsequently generated corrected paths, but the correction was rejected at the frozen proposal cap [1].

The original outcome remains INCOMPLETE_OR_STOPPED at the episode level. It produced no qualified observation, executed no real consumer and performed no scientific execution for a proposed design. A process exit code of zero therefore does not mean a successful E5 chain. The corrected call remains actual model output in the retained trace, rather than a researcher-written replacement.

All six actual provider requests have SETTLED usage records and no missing usage or unsettled reservation. The retained ledger records a charge upper bound of $0.0644115 and a cache-aware rate estimate of $0.035886444. These are pinned-rate accounting values, not provider invoice amounts. They belong to the original failed episode; the later adapter evaluation made zero additional provider requests. Host and authoring assistance are outside this episode's provider-usage total.

4 Separately authorized post-hoc qualification

The label `POSTHOC_HOST_ADAPTER_REEVALUATION` identifies a later engineering evaluation of the already generated corrected call. It introduces no new model proposal, new episode or proposer rerun. The review also replaces an over-specific paired-world-only condition with source dependence across legal design values; this changed qualification criterion is post hoc. The original protocol and failed outcome remain intact [5].

The corrected AST compares single-frame transmission energy with the current capacitor's ideal energy reserve between ready and off thresholds:

$$E_{\mathrm{TX}} = I_{\mathrm{TX}} \Delta t_{\mathrm{TX}} V_{\mathrm{out}} = 28.8\ \mu\mathrm{J}, \tag{1}$$

$$E_{\mathrm{reserve}} = \tfrac{1}{2} C (V_{\mathrm{ready}}^2 - V_{\mathrm{off}}^2) = 23.0\ \mu\mathrm{J}, \tag{2}$$

$$b = [E_{\mathrm{TX}} > E_{\mathrm{reserve}}] = \mathrm{true}. \tag{3}$$

The retained source uses $I_{\mathrm{TX}} = 8$ mA, $\Delta t_{\mathrm{TX}} = 2$ ms, $V_{\mathrm{out}} = 1.8$ V, $C = 10\ \mu$F, $V_{\mathrm{ready}} = 2.8$ V and $V_{\mathrm{off}} = 1.8$ V. Both sides have energy dimensions; the output is a dimensionless Boolean. The source-bound applicability constraints limit capacitance to $[4, 22]\ \mu$F, earliest transmission to $[0, 45]$ ms and transmission current to the record's 8 mA. Other device and waveform assumptions remain those of the frozen, uncalibrated ideal world.

S0 binds the acquired source hash and independently supplied unit map, then checks the expression with a separate Decimal AST interpreter. The value is true at 4 μF and false at 22 μF, rejecting a constant observation. The two transmission-time probes leave it true, as expected because the expression contains no time-scheduling field. Missing, self-signed and out-of-domain observations remain UNKNOWN; source hash binding passes [2]. These checks establish this source-dependent engineering relation under the stated assumptions. They do not calibrate the hardware.

The information added to the consumer is a derived relation among fields already acquired from the component record. It is construction, rather than selection of a predeclared observation. It is neither a new hardware measurement nor evidence that the model autonomously discovered a previously unseen physical relation. The energy comparison also omits startup work, losses and event ordering; its truth alone is not a complete service verdict.

5 Real consumer use and the strong static baseline

Without an admitted observation the consumer retains UNKNOWN and the original 10 μF, 45 ms design. After post-hoc qualification, Boolean true selects the model's original true branch: 15 μF with earliest transmission still 45 ms. The unused false branch proposes 10 μF and 30 ms; the receipts do not establish its service outcome. The consumer receipt binds the observation, unchanged model policy and complete executed input [3].

Design condition	C (μF)	Earliest TX (ms)	First RX (ms)	Service
Original design	10	45	–	FAIL
Post-hoc selected	15	45	46.9	PASS
Static complete information	10	0	5.6	PASS

Table 1: One public nominal BF task. The original exact-input result is reused; the selected design adds one full service execution. The strong static result is reused from its already executed input. All three have physical qualification UNKNOWN. A missing reception is shown as –.

The full service uses the unchanged BF feedback model and independent energy and receiver checks. The selected design receives one frame at 46.9 ms and passes complete service, whereas the original has no received frame and fails. Its independent energy ledger passes in both designs; energy-check success alone therefore cannot substitute for service success. The selected trajectory has zero brownouts before the deadline. The comparison is an executed post-hoc engineering consequence, not a prospectively admitted E5 benefit [4].

The strong static complete-information design also passes, using the original 10 μF capacitance and earliest transmission at 0 ms; its first reception is 5.6 ms. Thus this evidence neither requires capacitance growth nor demonstrates an advantage over complete static information. It also shows why the scalar energy flag cannot determine service independently of scheduling. The device record remains volatile. Reception before a reset, avoiding a reset and retaining pending state across a reset are different properties; the legacy `event_preserved_after_reset` label is not used here to claim the last.

6 Adapter repair, limits and reproducibility

The separate adapter-v2 regression receipt records two targeted regressions. For E5 it rejects the original malformed pointer before incrementing the valid proposal counter and admits the already generated corrected call as one valid proposal. It caps syntax rejections at two and retains the valid-proposal cap of one. The other regression concerns the H action contract and is outside this BF result. These are new-version, post-hoc qualification checks with no new model requests or scientific executions. They do not replace the original episode or establish a successful fresh live run [5].

The original episode and the adapter result answer different questions. The first demonstrates an actual failed attempt to use the interface. The second demonstrates that existing model output can be qualified, consumed

and linked to a nominal service change under revised host admission. There is one exposed development task, no randomized comparison, no held-out transfer evidence and no physical qualification. The strong static passing result precludes a superiority claim from this comparison. H proposal and design episodes are separate evidence and are not aggregated into this manuscript's outcome.

The accompanying `phase2-binding.json` records exact source paths, SHA-256 hashes, the frozen B interpreter, shared style, official vector English logo and generated TeX. `render_phase2.py` reads retained receipts and performs presentation arithmetic only; it invokes no provider, scientific solver or qualification interpreter. Building this manuscript adds no experimental evidence. The original failed episode, corrected model call, post-hoc plan, qualification, consumer, service and adapter regression records remain separately addressable. This is a local development manuscript, not a venue-compliance or hardware-acceptance result.

References

[1] S0. Original episode and usage receipts, 20 October 2026. `p2-e5-001`: `external/result.json`, `episode-result.json`, `provider-costs.json` and `pi-public-trace.jsonl`.

[2] S0. Post-hoc source and relation qualification. `p2-e5-001/posthoc-adapter-001/qualification.json`.

[3] S0. Post-hoc real consumer input and decision receipt. `p2-e5-001/posthoc-adapter-001/consumer.json`.

[4] S0. Original, selected and static-complete service receipts. `p2-e5-001/posthoc-adapter-001/service.json` and `result.json`.

[5] S0. Post-hoc validation plan and adapter-v2 targeted regressions. `docs/research/phase-plan-20261019/phase2/`, `posthoc-interface-validation-plan.json` and `adapter-v2/regression-result.json`.